Unit 10 - Week 8

Course outline

How does an NPTEL online course work?

Assignment Zero

Week 1

Week 2

Week 3

Week 4

Week 5

Week6

Week 7

Week 8

- Uncertainty in AI: Motivation
- Uncertainty in AI: Basics of Probability
- Uncertainty in AI: Conditional Independence & Bayes Rule
- Bayesian Networks: Syntax
- Bayesian Networks: Factorization
- Bayesian Networks: Conditional Independences and d-Separation
- Bayesian Networks: Inference using Variable Elimination
- Bayesian Networks: Reducing 3-SAT to Bayes Net
- Quiz : Assignment 8
- Week 8 Feedback Form
- New Lesson

Week 9

Week 10

Week 11

Week 12

Download Videos

Assignment Solution

Assignment 8

The due date for submitting this assignment has passed.
As per our records you have not submitted this assignment.

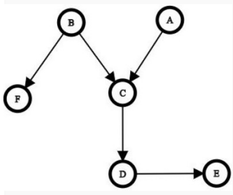
Due on 2020-03-25, 23:59 IST.

1) A box contains 4 blue and 2 red balls. We take one ball out of it, note down its color and put it back in the box. Thereafter we take out another ball and find it to be blue. Let the probability that the first ball was red given that the second ball is blue be p . Suppose instead of putting back the first ball in the box we threw it away. Now let the probability that the first ball was red given that the second ball is blue be q . What is the ratio of q/p . Round off your answer till 1 decimal point. (e.g. if your answer is 3.14159… then write 3.1)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 1.2

1 point

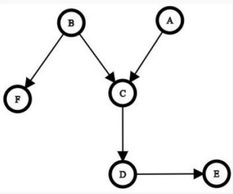
2) Consider the following Bayesian Network. Given evidence about C which of the following pair of variables are conditionally independent. *1 point*



- A and B
- A and D
- D and E
- A and F

No, the answer is incorrect.
Score: 0
Accepted Answers:
A and D

3) Consider the following Bayesian Network. Which of the following variables belong to Markov Blanket of F. *1 point*



- A
- B
- C
- D

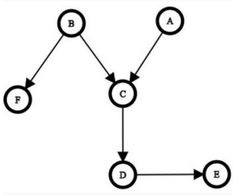
No, the answer is incorrect.
Score: 0
Accepted Answers:
B

4) Suppose A and B are conditionally independent given C. Then which of the following are true: *1 point*

- P(A,B) = P(A).P(B)
- P(A,B,C) = P(A|B,C).P(B|C).P(C)
- P(A,B,C) = P(A|C).P(B|C).P(C)
- P(A,B|C) = P(A|C).P(B|C)

No, the answer is incorrect.
Score: 0
Accepted Answers:
P(A,B,C) = P(A|B,C).P(B|C).P(C)
P(A,B,C) = P(A|C).P(B|C).P(C)
P(A,B|C) = P(A|C).P(B|C)

5) How many minimum parameters are required to represent the following Bayesian Network, if each variable is Boolean?



No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 12

1 point

6) There are two boxes A and B. Box A contains n red balls and 2 blue balls while box B contains 2 red balls and n blue balls. One of the two boxes is selected at random and two balls are drawn from it without replacement. If both the balls drawn are red, the probability of box A being selected is 6/7. Find the value of n.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 4

1 point

7) Find the difference in the minimum number of parameters required to specify a Bayesian network of 6 variables when there are no independencies and when all the variables are mutually independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 57

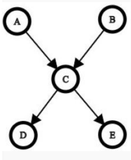
1 point

8) Which of the following statements is/are correct? *1 point*

- P(a | b, c) + P(¬a | b, c) = 1
- P(a | b, c) + P(a | ¬b, ¬c) = 1
- P(a | b, ¬c) + P(a | b, c) = P(a | b)
- P(a, b | c) = P(a | c).P(b | a, c)

No, the answer is incorrect.
Score: 0
Accepted Answers:
P(a | b, c) + P(¬a | b, c) = 1
P(a | b, ¬c) + P(a | b, c) = P(a | b)
P(a, b | c) = P(a | c).P(b | a, c)

9) Consider the following Bayesian Network with the given information. What is the value of P(¬a | d, e) up to 3 decimal places?

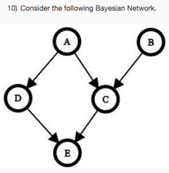


P(a) = 0.001, P(b) = 0.002
P(c | a, b) = 0.95, P(c | a, ¬b) = 0.94, P(c | ¬a, b) = 0.29, P(c | ¬a, ¬b) = 0.001
P(d | c) = 0.9, P(d | ¬c) = 0.05
P(e | c) = 0.7, P(e | ¬c) = 0.01

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 0.716

1 point

10) Consider the following Bayesian Network. *1 point*



Statement 1: B is independent of E given evidence at C.
Statement 2: B is independent of A given evidence at E.

- Both statements are correct.
- Both statements are incorrect.
- Statement 1 is correct and Statement 2 is incorrect.
- Statement 1 is incorrect and Statement 1 is correct.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Both statements are incorrect.